🧪 What Is Matter? Investigating Properties and States

Physical Science Unit, Lesson 1

📚 **Grade:** 2nd Grade ⏱️ **Duration:** 45–55 minutes 🔬 **Subject:** Physical Science — Matter
🧰 **Materials:** Household items only

Second graders act like scientists when they classify materials using evidence from what they observe.

📋 **NGSS Standard:**
2-PS1-1 — Plan and conduct an investigation to describe and classify different kinds of materials by their observable properties.

🎯 Learning Objectives

- Classify materials as solids, liquids, or gases using observable evidence
- Describe and compare matter using properties such as shape, texture, color, hardness, and ability to flow
- Explain how the same material can change state when conditions change, such as ice melting into liquid water

📦 Materials

🏠 What You Need (All Household Items)

- A small rock or pebble
- A cup of water
- An inflated balloon
- An ice cube
- A spoonful of cooking oil or syrup in a cup
- A plastic cup, bowl, or clear jar
- Paper and pencil or crayons

📖 Vocabulary

Matter
Anything that takes up space and has mass

Property
Something you can observe or test about a material

Classify
To sort things into groups using evidence

Solid
Matter that keeps its own shape

Liquid
Matter that flows and takes the shape of its container

Gas
Matter that spreads out to fill available space

🌍 The Science

Second graders are ready to do more than memorize the words solid, liquid, and gas. In this lesson, students act like scientists by observing materials closely, noticing their properties, and using evidence to decide how to classify them. Observable properties are things we can notice with our senses or simple tests, such as shape, texture, color, hardness, whether something pours, and whether it spreads out to fill a space.

Solids keep their own shape. Liquids do not keep their own shape — they flow and take the shape of the container they are in. Gases can be harder to notice because many gases are invisible, but they still take up space and can push on objects, like the air inside a balloon. Students should learn that classification is not guessing; it is using what you observe to support a decision.

This lesson also introduces the idea that matter can change state. An ice cube and liquid water are both matter, but they behave differently because one is a solid and the other is a liquid. That gives Grade 2 students a stronger bridge from simple sorting into real scientific thinking about materials and change.

> 💡 **Scientist tip:** Do not ask only *"What is it?"* Ask *"What evidence tells you that?"* That pushes the lesson from basic naming into real classification.

🔬 Investigation: Property Detectives

Step 1 — Observe like a scientist: Place the rock, water, balloon, ice cube, and oil or syrup on the table. Ask students to describe what they notice first: shape, feel, color, and whether the material stays put or moves.

Step 2 — Set up three groups: Label three spaces or papers SOLID, LIQUID, and GAS. Explain that scientists sort by evidence, not by guessing.

Step 3 — Test for shape: Pick up the rock and then the ice cube. Ask: *Do these keep their own shape when we move them?* Record the evidence and classify them as solids.

Step 4 — Test for flowing: Pour the water into a different container. Then slowly pour the oil or syrup. Ask: *What stayed the same, and what changed?* Students should notice that both liquids take the shape of the container, even if one flows faster than the other.

Step 5 — Test for gas evidence: Squeeze the balloon gently. Ask: *Can you see the gas clearly? What evidence shows it is there anyway?* Students should notice the balloon is full, takes up space, and pushes back.

Step 6 — Compare properties within a group: Ask students how the rock and ice cube are alike and different. Both are solids, but one may be rough while the other is smooth and cold. This helps students see that materials in the same state can still have different properties.

Step 7 — Watch a state change: Hold the ice cube for a minute or leave it on a plate. Ask: *What evidence shows the material is changing? Is it still keeping its own shape?* Students should notice that melting changes the solid into a liquid.

Step 8 — Make a claim with evidence: Have students finish a sentence such as: *"I know oil is a liquid because it poured and took the shape of the cup."*

💬 Discussion Questions

🗣️ **Question 1:** Why are both a rock and an ice cube solids even though they do not look or feel the same?

Suggested answer: They have different properties, but both keep their own shape when moved, which is the key evidence for classifying them as solids.

🗣️ **Question 2:** Water and oil are both liquids. What evidence shows that, and what properties make them different from each other?

Suggested answer: Both flow and take the shape of the container. They may differ in color, thickness, or how quickly they pour.

🗣 **Question 3:** If we cannot easily see air, why do scientists still say it is matter?

Suggested answer: Air takes up space, fills the balloon, and pushes back when the balloon is squeezed, so we have evidence that it is there.

🗣 **Question 4:** What happened when the ice cube started melting, and what does that tell us about matter?

Suggested answer: The ice stopped keeping its own shape and became liquid water. This shows the same material can change from one state of matter to another.

🔬 Student Worksheet — Matter Property Detective

Name: _____ Date: _____

Part 1 — Use Evidence to Classify

Write what you observed, then decide which state of matter fits best.

Material	What I observed	State of matter
Rock		
Water		
Balloon with air		
Ice cube		

Part 2 — Compare Two Materials

Rock and ice are both _____ because they both _____ .

Water and oil are both _____ because they both _____ .

Air in a balloon is a _____ because it _____ .

Part 3 — What Changed?

Watch an ice cube melt. Then explain the change.

At the start	After melting begins	What changed?

Part 4 — Scientist Sentence

Finish the sentence with evidence: "I know _____ is a _____ because _____."

👩‍🏫 Parent/Teacher Notes

- **Why this is stronger for Grade 2:** Students are not just naming states of matter. They are comparing materials, classifying by observable properties, and explaining their thinking with evidence.
- **Good evidence language to coach:** “It kept its shape.” “It poured.” “It spread out.” “It filled the balloon.” “It changed from solid to liquid.”
- **Important distinction:** A state of matter is not the same as a single property. A material can be smooth or rough and still be a solid.
- **Melting connection:** Keep the focus on behavior change. Ice and liquid water are the same material, but they behave differently in different states.
- **Strong student response example:** “I know the balloon has gas in it because the air filled the balloon and pushed back when I squeezed it.”
- **If students need support:** Give sentence frames, but still require a piece of evidence in every answer.
- **Extension:** Add another mystery material such as sand, honey, juice, or steam from warm water and ask students to defend their classification.

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🪨 Touch and Test: Texture, Hardness, and Flexibility

🧽 🪵

Physical Science Unit, Lesson 2

📚 **Grade:** 2nd Grade ⏱️ **Duration:** 45 minutes 🔬 **Subject:** Physical Science -- Material Properties
🧰 **Materials:** Household items only

Every material has unique properties -- and those properties determine what it is good for!

📋 **NGSS Standard:**
2-PS1-1 -- Plan and conduct an investigation to describe and classify different kinds of materials by their observable properties.

🎯 Learning Objectives

- Compare materials by observable properties: texture, hardness, and flexibility
- Record observations in a simple chart
- Connect material properties to real-world uses

📦 Materials

🏠 What You Need (All Household Items)

- Fabric scrap (any old shirt or cloth)
- A piece of aluminum foil (about 8 inches square)
- A kitchen sponge
- A rock or brick
- A small piece of wood or a wooden spoon
- A cotton ball
- A rubber band
- Paper and pencil for recording

📖 Vocabulary

Texture
How a surface feels: rough, smooth, bumpy, or scratchy

Hardness
How hard or soft a material is -- does it dent or scratch easily?

Flexibility
How easily a material bends without breaking

Rigid
Cannot bend easily -- stays stiff (like wood or rock)

Absorb
To soak up liquid -- like a sponge soaks up water

🌍 Background

Every material has properties we can observe with our senses. A property is a describing word that tells us something about what a material is like. Texture tells us how a surface feels -- is it rough and scratchy like sandpaper, or smooth like glass? Hardness tells us how resistant a material is to being dented or scratched. Try pressing your fingernail into a rock -- you will not leave a mark. Press it into a soft piece of foam -- you will!

Flexibility is another important property. A rubber band stretches and bends easily -- we call that flexible. A wooden board barely bends at all -- we call that rigid. Flexibility matters a lot in engineering. A building that is completely rigid might crack in an earthquake, while some flexibility helps it survive. That is why modern skyscrapers are designed to sway slightly in the wind!

Why do properties matter? Because different jobs need different materials. You would not build a house out of cotton balls (not hard enough). You would not make a sponge out of rock (not absorbent enough). By observing and comparing properties, we can make smart choices about which material to use for each job. Today you will test seven materials and record what you find.

🔬 Procedure: The Materials Properties Investigation

Step 1: Make a chart with these columns: Material | Texture (rough/smooth) | Hard or Soft | Flexible or Rigid | Other notes. You will fill in one row for each material.

Step 2: Pick up the fabric scrap. Run your fingers across it. Is it rough or smooth? Can you bend it easily? Try to scratch it with your fingernail. Record your findings.

Step 3: Test the aluminum foil. Scrunch it up into a ball. Flatten it back out. Is it rough or smooth? Hard or soft? Does it spring back or stay scrunched? Record.

Step 4: Squeeze the sponge. Press your finger into it. Scratch the surface with your nail. What do you find for texture, hardness, and flexibility?

Step 5: Test the rock. Try to bend it. Scratch it with your fingernail. Is it hard or soft? Rough or smooth? (The rock is very hard -- your nail will not leave a mark.)

Step 6: Test the piece of wood. Try to scratch it with your nail, try to bend it, and feel the surface. Record your observations.

Step 7: Squeeze and stretch the cotton ball. What texture does it have? Is it flexible or rigid? Is it heavy or light?

Step 8: Stretch the rubber band. Let it go. Bend it into a loop. Record your findings about texture, hardness, and flexibility.

Step 9: Look at your completed chart. Which material is the hardest? Which is the most flexible? Ask: which would you use to build a wall? Which would you use to clean a spill?

💬 Discussion Questions

🗣️ **Question 1:** Why would a builder choose wood over fabric for a wall?

Suggested answer: Wood is rigid and hard -- it holds its shape and supports weight. Fabric is soft and flexible -- it would not support a roof. Properties tell us what a material can and cannot do. This is why materials science is such an important field of engineering!

🗣️ **Question 2:** Why is rubber used for tires instead of wood?

Suggested answer: Rubber is flexible -- it absorbs bumps and shocks from the road. It also grips the road surface due to its texture. Wood would be rigid and would crack from repeated impacts, and it would not grip wet roads well. The properties of rubber make it perfect for tires.

🗣️ **Question 3:** What property would be most important for a raincoat? Why?

Suggested answer: Waterproof (non-absorbent) is most important -- you do not want rain soaking through. But flexibility matters too, since you need to move your arms. This shows that real-world material choices often need to balance several properties at the same time.

🗣️ **Question 4:** The aluminum foil is both flexible AND can hold its shape when crumpled. How is that possible?

Suggested answer: Metals like aluminum can be bent without snapping (flexible), but they do not spring back to their original shape the way rubber does. The foil takes a new shape and keeps it. This is different from a rubber band, which is elastic and snaps back. Advanced students can learn this is called "plastic deformation."

🔬 Student Worksheet -- Touch and Test

Name: _____ Date: _____

Part 1 -- Materials Properties Chart

Test each material and record what you find. Circle the correct word in each column.

Material	Texture	Hardness	Flexibility	Notes
Fabric	rough / smooth	hard / soft	flexible / rigid	
Foil	rough / smooth	hard / soft	flexible / rigid	
Sponge	rough / smooth	hard / soft	flexible / rigid	
Rock	rough / smooth	hard / soft	flexible / rigid	
Wood	rough / smooth	hard / soft	flexible / rigid	
Cotton Ball	rough / smooth	hard / soft	flexible / rigid	
Rubber Band	rough / smooth	hard / soft	flexible / rigid	

Part 2 -- Choose and Explain

Which material would you use to build a toy boat? Circle one: **wood / fabric / sponge / rubber band**

Why? _____

Part 3 -- Draw It!

✏️ *Draw the most flexible material you tested. Write its name below your drawing.*

Material name: _____

👩‍🏫 Parent/Teacher Notes

- **Hands-on observation is the heart of this lesson:** Do not rush the testing. Let students really press, bend, and scratch each material. The chart is secondary to the physical experience.
- **Hardness tip:** The fingernail scratch test works well for relative comparisons. A fingernail cannot scratch rock or wood but can dent clay or foam. This is actually a simplified version of the Mohs hardness scale used by geologists!
- **Flexibility vs. elasticity:** Rubber snaps back (elastic), foil stays crumpled (flexible but not elastic). Advanced students can explore this distinction. A rubber band is both flexible AND elastic.
- **For the chart:** Accept any reasonable descriptions. Texture is somewhat subjective -- rough to one person might be smooth to another. Focus on the process of careful observation, not on getting one "right" answer.
- **Real-world connection:** Bring out a raincoat, a sponge, and a wooden spoon and ask why each is made of the material it is. This grounds the lesson in everyday engineering decisions.
- **Boat building choice:** Wood is the best answer because it is rigid (holds shape), hard (will not compress), and (if sealed) waterproof. Students may also argue for the rubber band (waterproof, flexible) -- accept with discussion!

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🪣 Sink or Float? Testing Matter in Water 🍎🪙

Physical Science Unit, Lesson 3

📚 **Grade:** 2nd Grade ⏱️ **Duration:** 45 minutes 🔬 **Subject:** Physical Science -- Matter Properties
🧰 **Materials:** Household items only

Why does a tiny coin sink while a giant apple floats? Let's find out!

📋 **NGSS Standard:**
2-PS1-1 -- Plan and conduct an investigation to describe and classify different kinds of materials by their observable properties.

🎯 Learning Objectives

- Make predictions and test whether objects sink or float
- Record results and compare them to predictions
- Discover that shape (not just material) affects sinking and floating

📦 Materials

🏠 What You Need (All Household Items)

- A large bowl or pot filled with water
- An apple
- A coin (penny or quarter)
- A grape
- A wine bottle cork (or any natural cork piece)
- A wooden pencil
- A metal paper clip
- A small plastic toy (any small figurine or toy piece)
- A sheet of aluminum foil
- Paper and pencil for recording

📖 Vocabulary

Predict
To make a smart guess about what will happen before testing

Observe
To watch carefully and notice what actually happens

Float
To rest on top of water without sinking

Density
How much matter is packed into a space -- dense things tend to sink

🌍 Background

Have you ever wondered why a giant cruise ship floats but a tiny coin sinks? It seems backward! The secret is density. Density is how tightly packed the matter is inside an object. A coin is very dense -- lots of metal packed into a small space -- so it sinks. A cruise ship is mostly hollow inside, filled with air, so even though it is enormous, its overall density is lower than water, and it floats.

For 2nd graders, we can think of it this way: if an object is "heavier than water" for its size (meaning denser), it sinks. If it is "lighter than water" for its size (meaning less dense), it floats. Wood floats because wood has lots of tiny air pockets inside its fibers. Metal sinks because the atoms are packed very tightly together with no air gaps.

The most exciting discovery in this lesson is the aluminum foil experiment. A flat sheet of foil floats -- it traps air underneath and spreads its weight over a large area. But crumple the same foil into a tight ball, and it sinks! The amount of foil did not change, but the shape did. The flat sheet has more surface area touching the water, and the trapped air helps it float. The ball has no trapped air and is much denser in its compact form. Shape AND air both matter, not just what something is made of!

🔬 Procedure: The Sink or Float Investigation

Step 1: Fill your large bowl with water. Set all objects beside it. Create a prediction chart with columns: Object | Prediction (sink/float) | Result (sink/float) | Correct? (yes/no).

Step 2: Before dropping each object in, make a prediction and write it down. Remember: a prediction is not cheating -- it is scientific thinking!

Step 3: Gently place the apple in the water. Did it sink or float? Record the result. Compare to your prediction.

Step 4: Test the coin. Then the grape. Then the cork. Then the pencil. Then the paper clip. Then the plastic toy. Record each result.

Step 5 (The Exciting Part): Lay one sheet of aluminum foil flat and gently place it on the water surface. Does it float? Observe carefully.

Step 6: Take the foil out. Crumple it into a tight ball. Drop it in. What happens? Record your observation.

Step 7: Discuss: the foil is the exact same material both times. What changed? (The shape and the amount of trapped air changed!)

Step 8 (Bonus): Can you shape the foil into a boat shape that floats AND holds a small object on top? Try it!

🔑 **Key Discovery:** A flat sheet of foil floats because it has lots of surface area and traps air underneath. A crumpled foil ball sinks because it is compact and dense with no trapped air. SAME material, DIFFERENT behavior -- because the SHAPE changed!

💬 Discussion Questions

🗣️ **Question 1:** The foil ball sinks but the flat sheet floats. What changed?

Suggested answer: The amount of foil stayed the same, but the shape changed completely. The flat sheet traps air underneath it and spreads its weight over a large area. The crumpled ball has no trapped air and is very compact and dense. This shows that shape and air content matter for floating, not just what the material is made of.

🗣️ **Question 2:** Why does wood float but a metal coin sinks, even though the coin is much smaller?

Suggested answer: It is about density, not size. Wood has tiny air pockets inside its fibers that make it less dense than water overall. Metal atoms are packed very tightly, making it denser than water. Even a tiny metal coin sinks while a large wooden log floats!

🗣️ **Question 3:** How do ships float even though they are made of metal?

Suggested answer: Ships are hollow inside! They are mostly filled with air. Even though the walls are made of dense metal, the overall density of the ship (metal walls plus all the air inside) is less than water. Engineers design ships to be mostly empty inside on purpose -- that is what allows them to float despite being enormous.

🗣️ **Question 4:** What would happen if you put a ball of clay in the water? What if you shaped it into a bowl?

Suggested answer: A ball of clay sinks because it is dense with no trapped air. But if you flatten it into a bowl or boat shape, it can trap enough air to float! This is the same principle as the foil experiment -- the shape creates a pocket of air that helps the object ride on top of the water.

🔬 Student Worksheet -- Sink or Float?

Name: _____ Date: _____

Part 1 -- Predictions and Results

Write S (sink) or F (float) for your prediction, then test and record the result!

Object	My Prediction (S or F)	Result (S or F)	Was I right?
Apple			
Coin			
Grape			
Cork			
Wooden Pencil			
Paper Clip			
Plastic Toy			
Flat Foil Sheet			
Foil Ball			

Part 2 -- The Foil Experiment

The flat foil sheet floated because _____ .

The foil ball sank because _____ .

Part 3 -- Draw It!

✏️ *Draw the foil as a ball (sinking) and as a flat sheet (floating). Show the water level. Label each drawing.*

My most surprising result was: _____

👩‍🏫 Parent/Teacher Notes

- **The flat foil experiment is the highlight:** Make sure to do it! If the flat foil sinks immediately, it may be touching the bowl edge. Try laying it very gently flat in the center of the water surface.
- **Prediction charts:** It is completely okay to be wrong! Emphasize that scientists make predictions based on what they know, then adjust based on evidence. Wrong predictions are how we learn.
- **Density introduction:** The word "density" is introduced at a very conceptual level here. Do not expect mastery -- just plant the seed. The key phrase to use: "denser things tend to sink."
- **Paper clip trick:** A paper clip placed very gently and flat on the water surface can actually float due to surface tension (not density). If this happens, it is a great bonus observation! Dropping it at an angle will make it sink normally.
- **Apple note:** Most apples float because they have air pockets inside. Try cutting one in half and you can see air spaces in the flesh!
- **Missing cork substitution:** If you do not have a natural cork, a wine bottle plastic stopper, a small piece of styrofoam, or even a crayon will float similarly.
- **Connection to Lesson 6:** The foil boat bonus at the end previews the Unit 6 engineering challenge. Let students experiment freely here -- it builds excitement for the final project.

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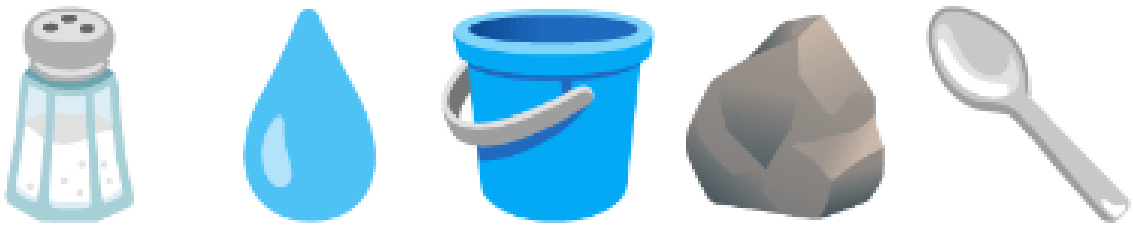
🧂 Mix It Up: Mixing and Separating Materials 💧🪣

Physical Science Unit, Lesson 4

📚 **Grade:** 2nd Grade ⏱️ **Duration:** 50 minutes 🔬 **Subject:** Physical Science -- Mixing and Separating
🧰 **Materials:** Household items only



Can you always get materials back after mixing them? Let's find out which changes are reversible!

📋 **NGSS Standard:**
2-PS1-3 -- Make observations to construct an evidence-based account of how an object made of a small set of pieces can be disassembled and put back together again.
2-PS1-4 -- Construct an argument with evidence that some changes caused by heating or cooling can be reversed and some cannot.

🎯 Learning Objectives

- Mix different materials and observe what happens
- Distinguish between mixtures that can be separated and those that are harder to separate
- Connect observations to the concept of reversible and irreversible changes

📦 Materials

🏠 What You Need (All Household Items)

- Table salt (about 2 tablespoons)
- Water (2 clear cups)
- Sand (about 2 tablespoons -- from outside, or a pinch of potting soil)
- Gravel or small pebbles (about 1/4 cup)
- A kitchen strainer or colander
- A spoon for stirring
- Paper and pencil for recording

📖 Vocabulary

Mixture
Two or more materials combined -- each material keeps its own properties inside

Dissolve
When a solid breaks into tiny invisible pieces in a liquid

Separate
To take a mixture apart and get the original materials back

Reversible
A change that can be undone -- you can go back to how it was

Irreversible
A change that cannot be undone -- no going back!

🌍 Background

A mixture is what you get when you combine two or more materials together. The interesting thing about most mixtures is that each material keeps its own properties even when mixed. Pour sand and gravel together -- the sand is still sand, the gravel is still gravel. They are just combined, not changed. With the right tool, you can get them apart again.

Some mixtures are easier to separate than others. Sand and gravel are easy to separate with a strainer because the gravel pieces are too big to fall through the holes, while the smaller sand particles fall through easily. Salt dissolved in water is trickier. When salt dissolves, it breaks into particles so tiny we cannot see them. You cannot filter them out with a regular strainer. But there is a way to get the salt back: evaporation. If you heat the saltwater (or let it sit in the sun), the water turns into vapor and escapes into the air, leaving the salt behind as white crystals.

Understanding mixtures and separation matters in real science and engineering. Water treatment plants separate dirt and bacteria from drinking water before it reaches your tap. Gold miners use flowing water to separate heavy gold flakes from lighter sand. Scientists in labs separate chemicals to identify what is in medicines or foods. Today, you will explore two classic mixing-and-separating challenges and decide which changes are reversible.

🔬 Procedure: Two Mixing Experiments

Step 1: Set up two clear cups. Label one "Salt Water" and one "Sand + Gravel."

Step 2 -- Experiment 1 (Salt + Water): Pour 1/2 cup of water into the first cup. Add 1 tablespoon of salt. Stir for 30 seconds. Watch carefully. Where did the salt go?

Step 3: Touch a clean finger to the surface of the water, then touch your finger to your tongue. What does it taste like? Is the salt gone, or is it still there but invisible?

Step 4: Record your observations: Did the salt disappear, or is it hiding inside the water? The salt DISSOLVED -- it is still there in tiny invisible pieces.

Step 5 -- Experiment 2 (Sand + Gravel): Mix the sand and gravel together in a cup. They are now a mixture. Look at it -- can you still see separate grains of sand and pebbles of gravel?

Step 6: Hold the strainer over a bowl. Pour the sand-and-gravel mixture into the strainer. What happens? The gravel stays in the strainer; the sand falls through!

Step 7: You just separated the mixture! Compare the two experiments: which was easy to separate, and which would be hard to separate?

Step 8: Discuss: Sand + gravel separation is reversible -- you got the same materials back. How would you get the salt back from the saltwater? (You would need to evaporate the water away by heating it!)

🔑 **Key Concept:** Sand + gravel: easy to separate with a strainer (reversible!). Salt + water: hard to separate -- the salt dissolved into tiny invisible pieces. You CAN get the salt back by evaporating the water, but it takes extra work!

💬 Discussion Questions

🗣️ **Question 1:** What would happen if you heated the salt water?

Suggested answer: If you heated the saltwater, the water would slowly evaporate, turning into water vapor. The salt cannot evaporate -- it would be left behind as white crystals on the bottom and sides of the cup. This is exactly how sea salt is made commercially! Ocean water is collected in large flat pools and the sun evaporates the water, leaving pure salt crystals behind.

🗣️ **Question 2:** Why could you separate the sand and gravel with a strainer, but not the salt and water?

Suggested answer: The strainer has holes. Gravel pieces are bigger than the holes, so they stay in the strainer. Sand particles are smaller and fall through. But salt that has dissolved in water breaks into particles so tiny that no regular strainer could catch them -- they are the size of individual molecules. To separate a dissolved solid from liquid, you need a different method: evaporation.

🗣️ **Question 3:** Is mixing salt into water reversible or irreversible?

Suggested answer: It is reversible, but tricky! You can get the salt back by evaporating the water. The salt does not change chemically when it dissolves -- it just breaks into very tiny particles that mix with water molecules. When the water leaves (evaporates), the salt particles come back together as crystals. Compare this to cooking an egg -- once cooked, you cannot uncook it. That is irreversible.

🗣️ **Question 4:** Can you think of a mixture from your kitchen that you could separate at home?

Suggested answer: So many options! Italian salad dressing (oil and vinegar separate when you stop shaking -- oil floats on top). Cereal and milk (pour off the milk, cereal stays). A bowl of mixed nuts (sort by size or type by hand). Orange juice with pulp (strain it to remove pulp). Frozen peas and corn mixed together (sort by color).

🔬 Student Worksheet -- Mix It Up!

Name: _____ Date: _____

Part 1 -- Experiment Observations

Experiment 1: Salt + Water

What did I do: _____

What I saw: _____

Reversible or irreversible? _____ Why? _____

Experiment 2: Sand + Gravel

What did I do: _____

What I saw: _____

Reversible or irreversible? _____ Why? _____

Part 2 -- Fill in the Blanks

When salt dissolves in water, it _____ but is still _____ .

A mixture is when two or more materials are _____ together.

Part 3 -- Sort the Changes

Circle the REVERSIBLE changes (changes you can undo):

Baking bread Mixing sand and gravel Cooking an egg

Melting ice Dissolving sugar in water Toasting bread

Part 4 -- Draw It!

✏️ *Draw what happened to the salt when you stirred it into water. Show the salt disappearing!*

👩‍🏫 Parent/Teacher Notes

- **Key NGSS concept:** This lesson introduces reversible vs. irreversible changes -- use these terms explicitly throughout the activity. Every time you do a step, ask: "Is this reversible?"
- **Salt water tasting:** Completely safe with plain table salt and clean water. This reinforces that the salt is still in the water even when invisible -- a powerful observation for kids who think it "disappeared."
- **Evaporation extension:** If time allows, put the saltwater cup on a sunny windowsill for several days and watch for salt crystals to form. This is a wonderful multi-day connection activity.
- **Sand sourcing:** Sand from a sandbox or garden works fine. It does not need to be clean play sand -- slightly natural, gritty sand actually makes the separation more dramatic.
- **Worksheet sorting answers (Reversible):** Mixing sand and gravel (strainer separates them), melting ice (refreeze it), dissolving sugar in water (evaporate to get sugar back). Irreversible: baking bread, cooking an egg, toasting bread (chemical changes -- new substances form).
- **Connection to real-world science:** Mention that water treatment plants, salt producers, and gold miners all use separation science. Making these connections helps students see science as relevant to the world around them.

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🧊 Hot and Cold: How Heating and Cooling Change Matter 🔥🍞

Physical Science Unit, Lesson 5

📚 **Grade:** 2nd Grade ⏱️ **Duration:** 45 minutes 🔬 **Subject:** Physical Science -- States and Changes
🧰 **Materials:** Household items only

Some changes you can undo. Others? No going back! Can you tell which is which?

📋 **NGSS Standard:**
2-PS1-3 -- Make observations to construct an evidence-based account of how an object made of a small set of pieces can be disassembled and put back together again.
2-PS1-4 -- Construct an argument with evidence that some changes caused by heating or cooling can be reversed and some cannot.

🎯 Learning Objectives

- Observe how heating and cooling change the state or appearance of matter
- Distinguish between reversible changes (melting, freezing) and irreversible changes (cooking, toasting)
- Build evidence-based arguments about why some changes can be undone and others cannot

📦 Materials

🏠 What You Need (All Household Items)

- A small pat of butter on a plate
- An ice cube in a cup
- A slice of bread
- A toaster (adult use only)
- Access to a freezer
- Optional: microwave (adult use) to demonstrate butter melting
- Paper and pencil for recording

⚠️ *Adult supervision required for toaster and any heating steps.*

📖 Vocabulary

Melt
Heating a solid until it becomes a liquid

Freeze
Cooling a liquid until it becomes a solid

Evaporate
Heating a liquid until it becomes a gas (steam or vapor)

Reversible
A change you can undo -- the material goes back to its original form

Irreversible
A change you cannot undo -- the material permanently changed

🌍 Background

Heat is a form of energy, and it has the power to change matter. When you add heat to a solid, you give its particles more energy, which can cause the solid to melt and become a liquid. Remove that heat by cooling, and the liquid can freeze back into a solid. Butter melts on a warm pan -- that is adding heat. Put the melted butter in the fridge and it hardens again -- that is removing heat. This change goes both ways, so we call it reversible.

Some changes go all the way. When you toast bread, the heat does not just warm the bread -- it causes a chemical change inside. New substances form, the sugars in the bread turn brown through a process called caramelization, and the texture becomes permanently crispy. No amount of cooling will bring back the soft, white interior. This is an irreversible change.

This distinction matters in everyday life. If ice cream melts, you can put it back in the freezer and it will refreeze (reversible, though the texture may change slightly). But if you burn toast or bake a cake, no amount of cooling will undo what heat has done (irreversible). Simple state changes like melting and freezing are usually reversible. Chemical changes caused by high heat -- cooking, baking, burning -- are usually irreversible.

🔬 Procedure: Three Heating and Cooling Experiments

Experiment 1 -- Butter: Place a small pat of butter on a warm plate (or have an adult briefly warm it in the microwave for 10-15 seconds on low). Watch what happens. Is the butter a solid or a liquid now?

Step 2: Put the melted butter in the fridge or set it outside in cold air. Wait 5-10 minutes. What happens? Did it go back to solid? Record your observation. This change is REVERSIBLE!

Experiment 2 -- Ice Cube: Place an ice cube in a cup. Watch it over 10-15 minutes (or hold it in your warm hands to speed things up). Record changes as it transforms.

Step 4: Observe the liquid water in the cup. Can you freeze it again? (Yes -- put it back in the freezer.) This change is REVERSIBLE! Water can go back and forth between solid and liquid many times.

Experiment 3 -- Toast: With an adult's help, toast a slice of bread in the toaster. Observe carefully: the color changes to brown, it smells different, and the texture changes from soft to crispy.

Step 6: Let the toast cool completely. Feel it. Did it become soft and white again? No! The change is IRREVERSIBLE -- the toast stays brown and crispy no matter how long you wait.

Step 7: Record all three experiments in a table: What happened / State changed? / Reversible or irreversible? / Your evidence (what you observed).

Step 8 (Discussion): When you bake a cake or boil an egg, are those changes reversible or irreversible? (Irreversible! Cooking always involves chemical changes that cannot be undone.)

🔑 **Key Rule:** Melting and freezing = REVERSIBLE (only the state changes, not the chemistry). Cooking, baking, and toasting = IRREVERSIBLE (the chemistry changes -- new substances form that you cannot uncreate).

💬 Discussion Questions

🗣️ **Question 1:** What other changes in cooking are irreversible?

Suggested answer: Baking a cake -- the batter is permanently transformed into cake. Boiling an egg -- the egg proteins change permanently and no cooling will make a raw egg again. Grilling a hamburger, caramelizing onions, baking cookies. Cooking almost always causes irreversible chemical changes. This is actually very useful -- we want cooked food to stay cooked!

🗣️ **Question 2:** Why can water freeze and melt over and over, but bread cannot be untoasted?

Suggested answer: Melting and freezing are physical changes -- only the state of matter changes, not the chemistry. Water molecules are still water molecules whether frozen or liquid. But toasting causes a chemical change -- new substances are actually created inside the bread. The chemistry cannot be reversed by simply removing the heat. New substances formed, and they are not the same as the original bread molecules.

🗣️ **Question 3:** Steam rises from a boiling pot of water. Is that steam gone forever?

Suggested answer: No! The steam is water in gas form. It can cool down and condense back into liquid water (you can see this on a cold window when steam hits it). This is the basis of the water cycle in nature. Water evaporates from oceans and lakes, rises as water vapor, cools in the atmosphere, and falls as rain or snow. Evaporation and condensation are reversible!

🗣️ **Question 4:** A chocolate bar melts in your hand. Is that reversible or irreversible? How do you know?

Suggested answer: Melting chocolate is reversible! If you chill it, it will harden back into solid chocolate. This is how chocolate candies are made: chocolate is melted, poured into molds, and then cooled to harden into the candy shape. However, once you eat the chocolate and your body digests it, THAT change is irreversible (digestion breaks chocolate down into completely different molecules).

🔬 Student Worksheet -- Hot and Cold Changes

Name: _____ Date: _____

Part 1 -- Experiment Results Table

Experiment	What Happened	State Changed?	Reversible or Irreversible?
🧈 **Butter**			
🧊 **Ice Cube**			
🍞 **Toast**			

Part 2 -- Reversible or Irreversible?

Write R (reversible) or I (irreversible) next to each change:

Melting ice _____ Baking a cake _____ Freezing juice into a popsicle _____

Toasting bread _____ Boiling water _____ Cooking an egg _____

Part 3 -- Fill in the Blanks

A _____ change can be undone.

An _____ change cannot be undone.

When butter melts, it changes from a _____ to a _____ .

Part 4 -- Draw It!

✏️ *Draw the butter before heating (solid) and after cooling back down (solid again). Label both pictures!*

Before heating (solid)	After cooling back down (solid again)

👩‍🏫 Parent/Teacher Notes

- **Adult supervision required:** Handle the toaster and microwave yourself. Have the child watch and observe carefully -- that is where the learning happens. Do not rush these steps.
- **Key vocabulary to reinforce:** Say "reversible" and "irreversible" constantly during each experiment. "Is this reversible? How do we know?" should be the recurring question throughout the lesson.
- **Butter experiment timing:** Butter melts quickly at room temperature on a warm day. For speed, a brief 10-15 second microwave burst on low works well. Be careful not to overheat -- you want melted butter, not splattered butter.
- **Ice cube observation:** If waiting for ice to melt fully takes too long, hold the ice in warm hands. It is a nice sensory experience too. The key is observing the state change from solid to liquid.
- **Sorting answers (R/I):** R = melting ice, freezing juice popsicle, boiling water (steam can re-condense). I = baking a cake, toasting bread, cooking an egg. Note: boiling water is reversible because steam condenses back to water.
- **Gifted extension:** Introduce the terms "physical change" (reversible, state only changes) and "chemical change" (irreversible, new substances form). This is 5th grade content, but advanced 2nd graders can grasp the core idea.
- **Connection forward:** This lesson builds directly toward the Lesson 6 engineering challenge -- students will apply their understanding of material properties to build a boat that uses foil's waterproof and flexible properties.

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🚤 Build the Best Boat: Engineering with Foil 🪙🔬

Physical Science Unit, Lesson 6 -- Engineering Challenge

📚 **Grade:** 2nd Grade ⏱️ **Duration:** 60 minutes 🔬 **Subject:** Physical Science -- Engineering Design
🧰 **Materials:** Household items only

This is the Unit 1 engineering challenge -- time to put everything you learned to work!

📋 **NGSS Standard:**
2-PS1-2 -- Analyze data obtained from testing different materials to determine which materials have the properties that are best suited for an intended purpose.
K-2-ETS1-1 -- Ask questions, make observations, and gather information about a situation people want to change to define a simple problem that can be solved through the development of a new or improved object or tool.

🎯 Learning Objectives

- Apply knowledge of material properties to make engineering design decisions
- Design, test, and improve a foil boat to hold the maximum number of pennies
- Practice the full engineering design process: design, build, test, analyze, improve

📦 Materials

🏠 What You Need (All Household Items)

- 3 sheets of aluminum foil (about 12x12 inches each -- or tear from a roll)
- 20 pennies (or other coins of equal size)
- A large bowl or pot of water
- A pencil and paper for sketching your design
- Optional: a ruler for measuring your boat

📖 Vocabulary

Engineer
A person who uses science and math to design and build things that solve problems

Design
A plan or drawing that shows what you want to build

Test
To try out your design and observe what happens

Improve
To make changes to your design to make it work better

Buoyancy
The upward force water pushes on an object -- what makes things float

🌍 Background

Engineers solve problems using science. Today you are going to be an engineer! Your challenge: design and build a boat from aluminum foil that can hold as many pennies as possible before sinking. This uses everything you have learned about matter. Foil is a solid with specific properties -- flexible, waterproof, and shapeable. Water creates an upward force on floating objects (called buoyancy). The shape of your boat determines how much weight it can support before water comes over the sides.

What makes a foil boat hold more pennies? Three things matter: width (a wider boat spreads the weight over more water), depth (taller walls keep water from spilling over the sides as pennies are added), and leak prevention (even a tiny gap or hole lets water in, and the boat will slowly sink). Good engineers think about all three before they start building.

This lesson is your chance to be a real engineer. You have three sheets of foil, so you have room to try multiple designs or iterate on one great idea. Keep track of how many pennies each design holds and what you changed between versions. Engineers actually learn more from failure than from getting it right the first time. If your boat sinks on the first test, that is not a setback -- that is data! It tells you exactly what to fix on the next try.

🏆 The Challenge

Goal: Build a foil boat that holds as many pennies as possible before sinking.

Rules: You may use only aluminum foil (no tape, no other materials). You must do at least TWO design iterations (try, improve, try again).

Success: Your best penny count after improving your design!

🔬 Procedure: Design, Build, Test, Improve!

Step 1 -- DESIGN (5 minutes): Before touching the foil, sketch your boat design on paper. Think about shape, wall height, and base size. Consider: wider or narrower? Tall walls or short walls? Flat bottom or pointed bow? Label your sketch.

Step 2 -- BUILD (10 minutes): Use your first sheet of foil to build your design. Take time folding the edges up to create walls. Make sure there are no gaps or holes at the corners -- those are where water sneaks in.

Step 3 -- TEST 1: Gently place your boat on the water. If it floats, start adding pennies one by one. Place them carefully in the center of the boat first. Count how many pennies it holds before sinking!

Step 4 -- RECORD: Write down your Test 1 penny count. Make a quick sketch of your boat design and note its shape, wall height, and any issues you noticed.

Step 5 -- ANALYZE: Why did your boat sink when it did? Were the walls too short (water came over the top)? Did it tip sideways (unstable, base too narrow)? Did it spring a leak at a corner? Each answer tells you what to fix.

Step 6 -- IMPROVE (5 minutes): Based on what you observed, plan one or two specific changes to your design. Write them down. What are you changing and why?

Step 7 -- BUILD 2: Use your second sheet of foil to build the improved design. Apply your analysis from Step 5 to make the changes you planned.

Step 8 -- TEST 2: Float your improved boat and load pennies. Record your Test 2 penny count. Compare to Test 1 -- did the improvement help?

Step 9 (Bonus): If you have a third sheet of foil and time, make a third iteration. What is your personal best penny record?

💡 **Penny loading tip:** Place pennies in the very center of the boat first, then work outward evenly. This keeps the boat balanced and prevents it from tipping to one side.

💬 Discussion Questions

🗣️ **Question 1:** What design features helped hold the most pennies? Why does a wider or deeper boat help?

Suggested answer: Width spreads the weight over more water surface, which increases buoyancy. Deeper (taller) walls keep water from splashing over the sides as pennies are added. A flat bottom also distributes the penny weight more evenly. The combination of wide base plus tall walls is usually the winning approach!

🗣️ **Question 2:** In Lesson 3, you learned that flat foil floats but a crumpled ball sinks. How did you use that information in your boat design?

Suggested answer: The flat sheet works because it has lots of surface area in contact with water and traps air underneath. The boat should be flat on the bottom (large area) and have walls to create an air pocket inside. Keeping the foil as flat and even as possible, not crumpled or leaky, applies the same principle that makes a flat foil sheet float.

🗣️ **Question 3:** If your first boat did not work well, how did you decide what to improve?

Suggested answer: You watch carefully when it fails -- that is the key engineering insight. Did the boat tip sideways? Then make the base wider for stability. Did water spill over the walls? Then make the walls taller. Did it spring a leak? Then fold the corners more carefully. Each failure tells you exactly what to fix. A failure is not a mistake -- it is information!

🗣️ **Question 4:** Could you build an even better boat if you used a different material? What properties would that material need?

Suggested answer: The material would need to be waterproof (water cannot pass through it), flexible enough to be shaped into a boat, and strong enough not to tear. Foil works because it is waterproof and very flexible, but it is thin and can tear at corners. A thicker waterproof material (like a plastic bag or heavy plastic sheet) might hold even more pennies. Engineers always ask "is this the best material for the job?"

🔬 Student Worksheet -- Build the Best Boat!

Name: _____ Date: _____

Step 1 -- My First Boat Design

✏️ *Draw your first boat design here. Label the shape, walls, and bottom.*

Test 1 Results:

Pennies held: _____ pennies

Why did it sink? _____

What I will change for Test 2:

_____

_____

Why I'm making this change:

_____

Step 2 -- My Improved Boat Design

✏️ *Draw your improved boat design. Show what you changed from Test 1.*

Test 2 Results:

Pennies held: _____ pennies

Did the improvement help? Yes / No

My Best Design Tip for Next Time:

_____

_____

_____

My best penny count was _____ pennies.

The most important design feature was _____ because _____ .

👩‍🏫 Parent/Teacher Notes

- **This is the unit culminating engineering challenge.** Encourage at least two full design-test-improve cycles. The iteration is the lesson -- not the penny count itself.
- **Penny loading tip:** Place pennies in the very center of the boat first, then work outward. This keeps the boat balanced. Dropping pennies from height or at an angle will cause splashing and premature sinking -- add them gently.
- **Common mistakes to watch for and discuss:** Gaps in foil corners (water seeps in), walls that are too low (water washes over), boat that is too narrow (tips sideways). Help students identify these through questioning -- do not just tell them the fix. Ask: "What do you notice? What might cause that?"
- **Reframing failure:** If a student gets frustrated after the first boat sinks, say: "That tells us so much -- now we know exactly what to fix. That is what engineers do." Model the language: "That's data, not failure."
- **Typical results:** First-attempt 2nd graders typically hold 5-15 pennies. With one thoughtful improvement (wider base or taller walls), 20-40+ pennies is very achievable.
- **Third sheet opportunity:** If time allows, the third sheet is a great bonus. Students who really understand the principles can sometimes hit 50+ pennies with a very wide, shallow boat with tall walls.
- **For younger siblings:** Let them add pennies and count along. They can participate meaningfully without building the boat themselves.
- **Connecting to unit vocabulary:** Ask students to use properties vocabulary: "The foil is _____ (flexible, waterproof) which makes it good for a boat because _____."

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🧪 Matter Is Everywhere

SOLID · LIQUID · GAS · PROPERTIES

SOLIDS

Keep their own shape

Hard, soft, rough, smooth

🪨 Rocks, wood, ice

Can hold, stack, break

LIQUIDS

Take container's shape

Flow and pour

🥛 Milk, water, juice

Hard to hold without a container

💨

🔍

⚠️ **NO SOLIDS**
Nothing to build with

⚠️ **NO LIQUIDS**
Can't drink or cook

⚠️ **NO GASES**
No air to breathe

⚠️ **NO PROPERTIES**
Can't describe the world

🌡️ Water can be solid, liquid AND gas on Earth

🪨 Diamonds are the hardest solid known

🫧 Soap bubbles are gas trapped in liquid film

🌊 Oceans hold 97% of Earth's water

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🧪 Matter Is Everywhere Quiz

Unit 1 -- Physical Science, Grade 2

📖 **Ask a grown-up to read the questions to you!**

⭐ **Score: 0 / 10**

🪨 **1** **Which of these is a SOLID?**

- Water in a cup
- Air in a balloon
- A wooden block
- Orange juice

💧 **2** **A liquid takes the _____ of its container.**

- Color
- Shape
- Weight
- Temperature

🎈

3 **Which state of matter fills ALL the available space in its container?**

- Solid
- Liquid
- Gas
- Ice

🔍

4 **What do we call a describing word for how a material looks or feels -- like "rough" or "hard"?**

- A mixture
- A property
- A gas
- A solid

🧽

5 **You can bend a rubber band easily and it snaps back. You cannot bend a rock at all. The rubber band is _____ and the rock is _____.**

- Flexible and rigid
- Rigid and flexible
- Hard and soft
- Heavy and light

6 **You drop a coin and a wooden pencil into a bowl of water. What happens?**

- Both sink
- Both float
- The coin sinks and the pencil floats
- The pencil sinks and the coin floats

7 **You stir salt into water and it seems to disappear. Where did the salt go?**

- It evaporated into the air
- It is still there, broken into tiny invisible pieces
- It turned into a gas
- It became a liquid

8 **You freeze a cup of water into an ice cube, then let it melt again. Is this change reversible or irreversible?**

- Irreversible, because ice looks different from water
- Reversible, because you can freeze and melt water over and over
- Irreversible, because cold changes water forever
- Reversible, but only once

9 **You toast a piece of bread. It turns brown and crispy. What kind of change is this?**

- Reversible -- you can make it soft again by cooling it
- Irreversible -- you cannot untoast bread
- Reversible -- bread always goes back to its original form
- Neither -- bread does not count as matter

10 **You build a foil boat but it sinks after only 3 pennies. What should an engineer do NEXT?**

- Give up and try a different project
- Improve the design based on what went wrong
- Build the exact same design again
- Use a coin instead of a penny

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